

SUPPLY CHAIN EXCEPTION CONTROL SYSTEM

Operational Application Specification for the Future Streamlit MVP

Version 1.0 | 20 July 2026

CONTROL POSITION

Candidate Risk is analytical, not a workflow state. The MVP permits only Open, Assigned, Investigating, Action Agreed, Monitoring, Resolved, Closed and Suppressed. Reopening, escalation, SLA breach and possible-resolution evidence are events or conditions. Resolved and Closed remain human-controlled.

Document field	Definition
Document status	Frozen UX and interaction baseline for implementation and acceptance; portfolio simulation, not validated company policy.
Prepared for	Supply Chain Exception Control System (SCECS) portfolio project.
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Primary operational grain	One purchase-order-line risk episode at one receiving site.
Operational time zone	Australia/Melbourne; business calendar Monday-Friday 08:30-17:00 excluding Victorian public holidays.
Scope	Operational application behaviour only. No Python, Streamlit code, SQL, Power BI measures, notifications or AI implementation.

Document authority and reading guide

Authority	Use
1. Project Charter v1.0	Governing business problem, objectives, scope, control principles, success measures and portfolio limitations.
2. Business Requirements Specification v1.0	Governing requirement and traceability identifiers; later frozen semantics supersede preliminary conflicts.
3. Frozen Operational Exception Lifecycle and Operating Model v1.0	Binding lifecycle, episode, ownership, SLA, approval, resolution and human-control semantics.
4. Logical Data Model and Data Contracts v1.0	Subordinate source for consistent field names, event evidence and run/score lineage.
5. Risk-Priority and Synthetic Data Specification v1.0	Subordinate source for risk explanation, missing-signal display, score bands, rule versions and receipt exposure.
6. This specification	Frozen application behaviour. Technical implementation may not silently redefine its controls.

Design commitments

- The interface prevents prohibited actions by role, lifecycle state, approval status and current record version. A failed validation causes zero domain mutation.
- Every displayed risk-priority score is an ordinal 0-100 control score, never a probability. Calculated score, calculated band and approved operational override remain separately visible.
- Every score snapshot preserves component availability, triggered/applied points, caps, missing signals, version, run, as-of time and source freshness.
- A lower score, zero residual quantity, receipt signal, supplier promise or possible-resolution flag never resolves or closes an episode automatically.
- Historical events, notes, approvals, evidence references, receipts and score snapshots cannot be edited or deleted through the application. Corrections append linked compensating events.
- Role switching is a simulation for portfolio demonstration and must never be described as production authentication, SSO or proven segregation of duties.
- The application exposes the latest successful publication separately from later failed run attempts so stale or failed data cannot masquerade as current.

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1. User-role catalogue

ID	Role	Accountability	Permitted interface scope	Key prohibitions
R01	Procurement analyst / buyer	Owns supplier-facing investigation, action planning, monitoring and normal resolution within authority.	Control Tower; Queue; Detail; own audit; approval requests.	Cannot self-approve material requests, change calculated scores, close, alter source facts, or administer runs.
R02	Inventory planner	Assesses item-site exposure and owns inventory actions such as substitution, reallocation and demand-date evidence.	Queue/detail; planner assessment; notes/evidence; inventory action/monitoring; category-authorised resolution.	Cannot approve material requests, change calculated scores/source facts, or close unless separately designated.
R03	Supply-chain manager	Owns queue control, assignments, escalations, material approvals, controlled resolution and normal closure.	All operational pages; team queue; approval inbox; audit; pipeline summary.	Cannot approve own material request; cannot edit history or perform data-administrator corrections by role alone.
R04	Data/system administrator	Monitors runs, defects, freshness, quarantine and corrected reprocessing; records attributable data corrections.	Pipeline Health; run detail; read-only operational detail and audit context.	Administrative access does not confer authority to investigate, suppress, resolve, close or approve operational risk.
R05	Governance / audit reviewer	Reviews control evidence and provides required concurrence for exceptional or critical material decisions.	Audit Review; evidence; approval inbox for concurrence; read-only queue/detail; pipeline assurance.	Cannot act as operational owner, silently correct events or approve a request they initiated.
R06	Read-only portfolio reviewer	Demonstrates traceability and limitations without making changes.	Read-only Control Tower, Queue, Detail, Pipeline Health and Audit Review; controlled exports.	No write actions, approvals, role/configuration changes or reprocessing.
SYS	System actor	Creates candidate evaluations/Open episodes, applies approved mappings, refreshes scores, records SLA events, expires suppressions and detects recurrence/evidence.	No interactive session. Events appear with actor type System and version context.	Cannot enter Resolved/Closed, approve, accept residual risk, delete history or silently reassign after mapping failure.

SIMULATED IDENTITY CONTROL

The application header must always show both the selected simulated user and role. Switching role starts a new simulated session context and appends no business event by itself. Every later write records the selected identity and role. Demonstrations must use distinct named identities for requester and approver; switching the same identity to another role does not satisfy separation of duties.

Role-resolution rules

- One current accountable owner is mandatory after assignment. Site, buyer group, supplier owner and planner are supporting dimensions, not parallel case owners.

- Automatic ownership precedence is supplier-site exception mapping; buyer-group plus site; product-category plus site fallback; staffed site duty queue; otherwise visible Open/Unassigned.
- Delegation is effective-dated and manager-approved; the base owner remains visible. Expiry does not silently reassign in-progress work unless the approval explicitly says so.
- Escalation adds manager accountability but does not change the owner or lifecycle state. Reassignment is a separate controlled action and never resets SLA.

2. Application navigation

Navigation area	Entries	Behaviour
Global header	SCECS name; SYNTHETIC DATA label; selected simulated identity/role; latest successful publication; current time zone; Help/Control Legend.	Persists on every page. Stale/failed status is text plus icon, not colour only. Role switch requires an explicit session change.
Primary navigation	Control Tower; Exception Queue; My Approvals; Pipeline Health; Audit Review.	Items remain consistently ordered. Access-denied pages are not rendered; deep links are re-authorised on load.
Context navigation	Queue breadcrumb; Previous/Next in current result set; exception tabs; Back to queue.	Preserves queue filters, sorting and page. Never relies on browser Back as the only return path.
Action rail	State- and role-eligible actions for the current exception.	Unavailable actions are absent or disabled with a non-sensitive reason. Server-side rules repeat every check on submit.
Global banners	Latest run failed; published data stale; section unavailable; unsaved form; record changed; simulated-security limitation.	Banners identify consequence and safe next action. A failure never changes the published-current pointer.

Recommended page routes

Page ID	Route label	Primary users	Purpose
P01	Control Tower	Buyer, planner, manager, reviewer	Operational starting point: workload, urgent conditions, pending approvals and data confidence.
P02	Exception Queue	Buyer, planner, manager, reviewer	Ranked, filterable worklist with stable sorting and drill-through.
P03	Exception Detail	All roles, with role-specific actions	Single episode workspace for facts, risk explanation, work, evidence, approvals, SLA and immutable history.

Page ID	Route label	Primary users	Purpose
P04	My Approvals	Manager; governance reviewer	Pending, conditional, approved, rejected, expired and withdrawn material requests.
P05	Pipeline Health	Administrator; manager; audit; reviewer	Current publication, run attempts, data freshness, validation, quarantine, reconciliation and coverage.
P06	Audit Review	Governance; manager; reviewer; administrator	Cross-episode event search and evidence export without mutation.
P07	Help and Control Legend	All	State definitions, priority/SLA/missing-data legend, role-simulation warning and application limitations.

3. Screen inventory

ID	Screen	Type	Minimum content
S00	Application shell	All pages	Identity/role, simulation label, latest successful run, stale/failure state, navigation and time zone.
S01	Role switch panel	Demonstration only	Select a named simulated identity-role pairing; display non-production authentication limitation.
S02	Control Tower	Operational overview	Actionable counts and watchlists with direct filtered drill-through; not a replacement for Power BI.
S03	Exception Queue	Worklist	Columns, filters, documented ranking, SLA/ownership/missing-data indicators, pagination and drill-through.
S04	Exception Detail - Summary	Episode orientation	Identity, state, priority, score, owner, SLA, next action, key exposure and eligible actions.
S05	Exception Detail - Risk & Source	Explainability	Score contributions, missing signals, source hierarchy/freshness, inventory/demand exposure, schedules and receipts.
S06	Exception Detail - Work	Case handling	Assignment, investigation notes, action agreement, monitoring checkpoints, evidence and possible-resolution review.
S07	Exception Detail - Approvals	Authority	Requests, decisions, conditions, expiry, execution state and separation-of-duties result.
S08	Exception Detail - SLA & Audit	Control evidence	Obligations, pauses, reminders, breaches, escalations, lifecycle/ownership history and event detail/export.
S09	Assignment / reassignment form	Controlled form	Select eligible owner, reason, handover and approval reference where required.
S10	Investigation form	Controlled form	Explicit work acceptance, validity assessment, cause/status, reviewed facts, next review and evidence.
S11	Action agreement form	Controlled form	Complete plan, accountable actor, commitment, due date, residual exposure, evidence, approvals and monitoring condition.

ID	Screen	Type	Minimum content
S12	Monitoring forms	Controlled form	Initiation, measurable condition, expected event/date, checkpoints, evidence and return-to-investigation decision.
S13	Resolution form	Controlled form	Ten categories with conditional evidence, residual risk, approvals, monitoring result and closure-review date.
S14	Closure form	Controlled form	Resolved-only verification checklist, independent reviewer rule, final outcome/SLA, limitations and approvals.
S15	Suppression request/apply form	Controlled form	Reason, evidence, expiry, scope, recurrence criteria, approval and separate execution.
S16	Approval inbox/detail	Controlled form	Request context, authority, evidence, independence, decision, conditions and expiry.
S17	Pipeline Health	Operational assurance	Published-current record and immutable attempts, steps, defects, quarantine, reconciliation, coverage and freshness.
S18	Audit Review / event detail	Assurance	Filters, canonical event payload, causation/correlation, before/after, evidence and version references.
S19	Error/unauthorised/not found	Safe failure	No data leakage; stable return route; correlation reference for support.

Control Tower content

Region	Required fields / measures	Drill-through
Publication strip	Latest successful run ID; source as-of; published-at; dataset age; run status; failed attempts since publication; validation confidence.	Pipeline Health filtered to publication/attempt.
Workload cards	Active excluding Suppressed; Critical; High; Open-Unassigned; breached assignment; breached action/monitoring; Resolved awaiting closure; Suppressed.	Queue with fixed filter chips displayed.
My work	Owned active count; next action due today; overdue checkpoints; possible-resolution evidence awaiting review.	Queue filtered to selected identity.

Region	Required fields / measures	Drill-through
Manager watchlist	Unassigned Critical/High; breached Critical/High; approvals pending; suppression expiring; exceptional closure requests.	Queue or Approvals with documented sort.
Data confidence	Freshness by PO/receipt/inventory/demand/confirmation/supplier performance; missing-signal-qualified count; invalid/quarantined count.	Pipeline Health or Queue filtered to missing signal.
Operational caution	Counts are based on latest successful publication plus subsequent workflow events; analytical trends and success claims belong in Power BI/report evidence.	Help/Control Legend.

4. Text-based wireframes

P01 - Control Tower

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| SCECS | SYNTHETIC DATA | User: Priya Shah | Role: Supply-chain manager (SIMULATED) |

| Published current: RUN-20260720-01 | As-of 20 Jul 08:00 AEST | Age 6h | Status: CURRENT |

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| Control Tower | Exception Queue | My Approvals (7) | Pipeline Health | Audit Review |

+-----+

| Active 184 | Critical 18 | High 61 | Unassigned 9 | SLA breached 14 | Awaiting closure 11 |

| Suppressed 6 | Possible resolution awaiting review 8 | Latest failed attempts 1 |

+-----+

| MY WORK | MANAGER WATCHLIST |

| 4 overdue checkpoints [Open filtered queue]| C-1029 Critical / unassigned / 42m remaining |

| 3 possible resolutions [Review] | C-0981 Critical / action SLA breached |

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| DATA CONFIDENCE: Inventory 6h | Demand 6h | Receipts 2h | Qualified missing signals: 12 |

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P02 - Exception Queue

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| Exception Queue | 184 results | Published as-of 20 Jul 08:00 AEST | [Saved view: My active] |

| Search [PO, item, supplier, exception] Owner [Mine] State [Active] Priority [All] SLA [All] |

| Site [All] Supplier [All] Need date [-- to --] Missing signal [All] [Apply] [Clear] |

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| Priority | SLA | Exception | State | Owner | Site | Supplier / PO-line | Need | Next action |

| CRIT 82 | BREACHED Action | EXC-00129 | Investigating | Priya | MEL | S-044 / PO-881-2 ... |

| HIGH 58 | 1h 42m Assign | EXC-00144 | Open | UNASSIGNED | SYD | S-020 / ... |

| MED 42 | PAUSED | EXC-00118 | Suppressed | Alex | MEL | ... | Missing: ASN |

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| Showing 1-50 of 184 | Rows [50] | Previous | 1 2 3 4 | Next | Sort: Control priority |

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P03 - Exception Detail

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| Queue > EXC-00129 | State: Investigating | Effective: CRITICAL | Score: 82 (not probability) |

| Owner: Priya Shah | Site: Melbourne | SLA: ACTION PLAN BREACHED | Next action: supplier call |

| [Start/record action] [Reassign] [Request suppression] [Resolve] (eligible actions only) |

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| Summary | Risk & Source | Receipts | Work | Approvals | SLA | Audit |

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| WHY PRIORITISED          | CURRENT EXPOSURE          | | | |
| Overdue residual +30     | Residual: 80 EA / A$165,000 |
| Projected stockout +25   | Need: 22 Jul | Expected usable receipt: 26 Jul |
| Critical item +5         | Projected shortfall: 35 EA on 23 Jul |
| Supplier OTIF +15; cap applied -? | Source freshness: Qualified (ASN unavailable) |
| Total 82 | RPR-1.0.0 | RUN-... | as-of ... | Possible resolution: NONE |
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| Latest work / evidence / approvals / timeline; historical entries are read-only |
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P04 - My Approvals

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| My Approvals | Pending 7 | Conditional 2 | Approved-not-applied 1 | Expiring 3 |
| Type [All] Priority [Critical/High] Site [All] Requester [All] Age [Oldest first] |
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| Request | Exception | Type | Original -> Requested | Requester | Evidence | Age | Action |
| APR-119 | EXC-129 | Suppression | Active -> 1d | Priya | 2 refs | 42m | [Review] |
+-----+
| Decision panel: independence PASS | authority PASS | evidence 2/2 | conditions [ ] |
| [Approve] [Approve with conditions] [Reject] Rationale [required] Expiry [conditional] |
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P05 - Pipeline Health

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| Pipeline Health | PUBLISHED CURRENT: RUN-20260720-01 | SUCCESS | age 6h |
| Latest attempt: RUN-20260720-02 | FAILED validation | NOT PUBLISHED | current data unchanged |
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| Source freshness | PO 6h OK | Receipt 2h OK | Inventory 6h OK | Demand 6h OK | ASN 29h STALE |
| Control totals | Source 50,128 | Accepted 50,011 | Quarantined 117 | Reconciled YES |
| Coverage | Eligible 1,502 | Scored 1,502 | Qualified 48 | Invalid 0 | Opened 27 |
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| Run attempts: ID | start/end | step | status | defects | duration | publication decision [View] |
| Validation defects: severity | code | entity | count | treatment | owner | evidence [View] |
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5. Required fields by screen

Screen	Required fields and content
S00 Application shell	Synthetic-data environment label; selected user ID/name; simulated role; latest successful run; source as-of; published-at; age/freshness status; failed attempts since publication; Australia/Melbourne label; app release.
S01 Role switch	Named simulated identity; eligible role; role description; limitation acknowledgement; switch action. Do not allow free-text identity or role combination that defeats distinct-person checks.
S02 Control Tower	Current publication reference; active counts by state/priority; unassigned/breached/pending approval/expiring suppression/possible-resolution counts; my work; manager watchlist; source freshness summary; fixed drill filters.
S03 Exception Queue	Result count; active filters; documented sort; data as-of; pagination; row fields defined in Section 5.1; clear empty/filter state.
S04 Detail Summary	Exception/episode ID; predecessor/successor; PO line/site; current state; opened/age; calculated score/band; effective operational priority/override; owner/base owner/delegate; SLA summary; next action; residual quantity/value; need/receipt/stockout dates; missing-signal status; eligible actions.
S05 Risk & Source	Candidate snapshot selector; evaluation/run/rule/as-of/fingerprint; component rows; caps/floor/total; explanation codes; missing signals/reasons/freshness; source PO/supplier/item/site/dates; inventory/demand projection; schedules; receipts/allocations/corrections.
S06 Work	Ownership history; investigation entries; supplier/planner contact; action plans/amendments; monitoring checkpoints; evidence references; possible-resolution evidence and review disposition.
S07 Approvals	Request ID/type/status; requester/time; original/requested values; evidence; required roles; decision(s); conditions; expiry; SoD result; applied/not-applied/expired effect.
S08 SLA & Audit	All obligations; original/current due; elapsed/paused/remaining business minutes; calendar version; reminders/breaches/escalations; event sequence, actor, Melbourne time, before/after, reason, source, causation/correlation, evidence/approval/version references.
S16 Approval inbox	Counts by status; request age; type; exception/priority/site; requester; required approver/concurrence; evidence completeness; conflict/SoD status; decision form.
S17 Pipeline Health	Published-current run; latest attempt; run/step IDs; start/end/duration/status; source loads/as-of/age; validation defects; quarantine; control totals; eligible/scored/opened coverage; publication decision; rule/data contract versions; checksums/evidence links.
S18 Audit Review	Filters by exception/event/actor/role/date/type/run/rule/approval; total results; immutable event summary; event detail; export scope, generation time and publication reference.

Screen	Required fields and content
S19 Error page	Failure class; plain-language consequence; correlation/reference ID; safe return destination; retry only when idempotent and safe; no stack trace or secret.

5.1 Exception Queue specification

Column	Visibility	Definition
Priority	Default	Effective band text chip; score; calculated band when overridden; criticality and override markers. Never show percent or probability gauge.
SLA	Default	Most urgent active obligation; BREACHED/PAUSED/DUE text; business time remaining/overdue; Melbourne due time; additional-clock count.
Exception ID	Default	Stable episode identifier and predecessor marker; selectable drill-through link.
State	Default	One of the eight states only. Escalation, breach and reopened appear as adjacent event badges, never status values.
Owner	Default	Current named owner; UNASSIGNED in text; base/delegate marker; reassignment pending marker.
Site	Default	Receiving site code/name.
Supplier	Default	Supplier code and short synthetic name.
PO / line	Default	Stable PO and line identifier; split schedules shown as a count, not separate cases.
Item	Default	Item code, short description and criticality label.
Exposure	Default	Residual quantity and UOM; residual AUD value; projected shortfall where present.
Key dates	Default	Need date; expected usable receipt; earliest projected stockout. Unknown values explicitly display Missing/Unavailable.
Risk reasons	Default	Top two ordered explanation labels plus '+n'; details on drill-through.

Column	Visibility	Definition
Next action	Default	Next-action/checkpoint label and date; OVERDUE text when applicable.
Missing data	Default	Complete / Qualified / Invalid text and number of unavailable/invalid signals.
Updated	Default	Latest score as-of and latest workflow event time; distinguish analytical freshness from human event recency.
Buyer group/category	Optional	Ownership routing context.
Supplier OTIF/sufficiency	Optional	Trailing metric and n, or insufficient-history label.
Approval/suppression	Optional	Pending approval type; suppression expiry/remaining time.
Opened/age	Optional	Opened time and business/calendar age.
Run/rule	Optional	Current candidate evaluation, run and rule version.

Filters

Filter group	Values / behaviour
Search	Exact/contains search across exception ID, PO, line, item code/name and supplier code/name. Trim whitespace; minimum 2 characters except exact ID; no wildcard syntax in MVP.
Ownership	Mine; unassigned; my site; delegated to me; specific current owner; base owner; buyer group. Default is role-specific but always displayed.
Lifecycle	Open, Assigned, Investigating, Action Agreed, Monitoring, Resolved, Suppressed; Closed only when Include historical is selected.
Priority	Effective band; calculated band; calculated-critical-with-override; item criticality; score range 0-100.
SLA/action	Breached; due within 1h/4h/today; paused; assignment/action/monitoring/closure obligation; next action overdue; escalation present.
Operational dimensions	Site; supplier; buyer group; product category; item; resolution category where applicable.
Dates/age	Need date range; expected receipt range; stockout date range; opened date/age band; next action date; suppression expiry.
Data confidence	Complete; Qualified missing signal; Invalid; named missing-signal code; freshness status; possible-resolution evidence present.

Filter group	Values / behaviour
Authority	Approval pending; approved-not-applied; rejected; expired; suppression status/reason.

Sorting and stable priority

- Assign a control tier: Critical with active breach; Critical unassigned; High with active breach; High unassigned; any other breach or overdue next action; all remaining active work; Resolved awaiting closure; Suppressed; Closed when included.
- Within a tier, sort by effective operational severity descending. If an approved reduction exists, retain a Calculated Critical/High marker and allow an explicit calculated-band filter.
- Then sort by earliest non-null projected stockout date, then need date, with unavailable dates after known dates but carrying a missing-data marker.
- Then sort approved item criticality descending, longest business time since last meaningful action, oldest opened-at, and finally exception ID ascending as the stable tie-breaker.
- User sorting is permitted on one primary column plus stable exception-ID tie-breaker. The UI shows when the documented control ranking has been replaced and provides Restore control ranking.

Indicators, pagination and drill-through

Control	Frozen behaviour
Priority	Use labelled chips Critical/High/Medium/Monitor plus numeric score. Effective and calculated values are both visible when different. Do not rely on red/amber/green alone.
SLA	Show obligation name, exact business time remaining/overdue, Melbourne due time and PAUSED/BREACHED text. An already recorded breach remains visible after extension or reassignment.
Ownership	Show named current owner or UNASSIGNED. Delegated cases show 'Delegate (base: owner)'. Escalation shows manager-accountability badge without changing owner.
Missing data	Show Complete, Qualified or Invalid. Tooltip/detail lists signal code, source age, allowed freshness and affected rules. Missing never renders as 0, None or blank.
Pagination	Server-side pagination; default 50 rows; options 25, 50 and 100. Show '1-50 of N', first/previous/next/last, current page and total pages. Changing filters returns to page 1; changing page retains filters/sort.
Drill-through	Select Exception ID or row Open control. Open in same app context; preserve query/filter/sort/page state; show breadcrumb and Previous/Next within the frozen result set. Re-authorise direct links.
No bulk mutation	The MVP has no bulk assignment, suppression, resolution or closure. Each controlled decision requires episode-specific context and evidence.

5.2 Exception Detail specification

Tab	Section	Required content
Summary	Identity and state	Episode/predecessor/successor; PO-line/site; state; calculated/effective priority; score; owner; SLAs; next action; exposure; missing-data status; possible-resolution banner; eligible actions.
Risk & Source	Risk explanation	Snapshot selector; rule/component/availability/observed value/threshold/gross/applied points/cap; subtotal/floor/total; explanation codes; score confidence; missing signals; rule/run/as-of/fingerprint.
Risk & Source	Source data	PO header/line and version; supplier; item/site; ordered/residual quantities; price/currency/AUD value; requested/need/confirmed/expected usable-receipt hierarchy; freshness and quarantine flags.
Risk & Source	Inventory and demand	Snapshot time/age; on-hand/available/safety stock; daily projected available balance horizon; earliest shortfall; demand requirement set and freshness; substitutions/protected stock evidence.
Receipts	Schedules	Schedule ID; due/need date; ordered/residual quantity; status; earliest threatened flag. Schedules remain within one episode.
Receipts	Receipt history	Receipt event; physical/usable/rejected/quarantined quantity; UOM; allocation to schedule; posted time; source; correction/reversal link; net residual after event; no overwritten cumulative history.
Work	Ownership history	Base/current owner; delegate; assignment basis/mapping version; before/after; reason; actor/time; handover; approval; SLA effect=no reset.
Work	Investigation	Append-only notes; validity assessment; cause hypothesis; contacts; reviewed facts; next review; evidence; residual exposure at event time.
Work	Action and evidence	Action category; rationale; owner; commitment/target; dates/quantities; next action; evidence; approvals; residual risk; amendments as new events.
Work	Monitoring	Initiation; measurable condition; expected event/date; checkpoints; reminders/breaches; current result; next review; return-to-investigation reason.

Tab	Section	Required content
Approvals	Requests and decisions	Original/requested values; requester; required roles; decisions; conditions; expiry; SoD; executed effect; expired/revoked state.
SLA	Obligations/events	Assignment, action, monitoring and closure-review clocks; original/revised due; pause intervals; elapsed/remaining; reminders; breach and escalation events; calendar version.
Audit	Lifecycle timeline	Chronological Open/transition/reopened/suppression events plus same-state score, ownership, note, evidence, approval, breach and correction events.
Audit	Event detail	Event/sequence; actor type/user/role; effective/recorded/Melbourne time; prior/new; reason/source; correlation/causation; evidence/approval/run/rule/app version.
Summary/Work	Possible-resolution evidence	Detected source event, suggested category, quantity/date/source, confidence/data warnings, reversal risk and review status. Human may dismiss, investigate or open Resolution form; the flag never changes state.

ACTION GATING

The action rail is computed from current lifecycle state, selected identity, role, ownership/authority, approval status, evidence completeness and record version. Client-side hiding is usability only; the same rules must be rechecked at submit. If any check fails, the application records no domain change and returns a specific message.

6. Allowed actions by role and state

State	Allowed actions	Authority	Prevented actions
Open	Assign; triage note; request suppression; link source correction; manager escalation note.	Analyst/manager assign; analyst/planner note; manager requests but cannot self-approve; admin links correction; system opens/escalates.	Direct Investigating, Action Agreed, Resolved or Closed; silent default owner; suppression without approved request.
Assigned	Begin investigation; reassign/delegate; note/evidence; request suppression; risk refresh by system.	Current owner or manager begins; analyst/manager reassign; planner contributes; manager approves requests when independent.	System begins investigation; direct action/monitoring/resolution/closure; SLA reset on reassignment.
Investigating	Notes/contacts/evidence; planner assessment; agree action; rapid resolution; reassign; request suppression.	Owner/manager lead; planner contributes and resolves inventory categories; manager controlled routes.	Incomplete Action Agreed; low-score resolution; unresolved mandatory action; self-approved material decision.
Action Agreed	Confirm initiation; amend plan as new event; seek approval; return to Investigating; immediate evidenced resolution; request suppression.	Action owner/manager; planner for inventory actions.	Overwrite plan; Monitoring without measurable condition/evidence; automatic resolution; indefinite suppression.
Monitoring	Checkpoint; next review; escalation; return to Investigating; submit resolution; reassign; request suppression.	Owner/planner/manager within authority; system records reminders, breaches and possible evidence only.	Receipt flag resolves automatically; delete failed commitment; suppress solely to hide breach.
Resolved	Review evidence; append monitoring result; withdraw resolution to Investigating; close.	Resolver/manager may withdraw; authorised manager/designated reviewer closes; governance concurs exceptional closure.	Edit resolution; direct suppression; system closes; self-review where independent approval is required.
Suppressed	Review; revoke; request extension; inspect scores; automatic expiry/reactivation.	Manager/governance decisions; system executes expiry; operational users view and add evidence only.	Resolve/close from Suppressed; edit expiry; overlapping interval; remain past expiry; SLA reset.
Closed	View/export; audit annotation; system opens linked successor on material recurrence.	All authorised viewers; governance annotation; system successor opening.	Edit/reopen predecessor state; add operational work; delete history; unlinked duplicate successor.

Transition matrix

From	To	Actor	Required gate	Evidence/event
No episode	Open	System	Qualifying candidate; no active duplicate; no blocking candidate control.	OPENED; score/run/SLA events atomically.
Open	Assigned	System, analyst or manager	Valid eligible owner; mapping basis or manual reason; Critical manual change approval where applicable.	ASSIGNED; ownership before/after; assignment SLA stop.

From	To	Actor	Required gate	Evidence/event
Open/Assigned/Investigating/Action Agreed/Monitoring	Suppressed	Manager-approved executor	Approved request, distinct approver, evidence, expiry, conditions and concurrence where required.	SUPPRESSION_APPLIED; prior state/owner and SLA pause.
Assigned	Investigating	Owner or manager	Explicit begin-work confirmation and complete investigation-start fields.	INVESTIGATION_STARTED.
Investigating	Action Agreed	Owner; manager for controlled plan	Complete plan, evidence, monitoring expectation and approvals.	ACTION_AGREED; action-plan SLA result.
Investigating	Resolved	Authorised human	Rapid objective/category evidence, no unresolved actions, residual-risk control and approvals.	RESOLVED; possible-evidence review linked if used.
Action Agreed	Monitoring	Action owner	Initiation evidence and measurable expected condition/date.	MONITORING_STARTED; monitoring obligation.
Action Agreed	Investigating	Owner or manager	Plan withdrawn/rejected/invalidated with reason and evidence.	ACTION_WITHDRAWN; prior plan retained.
Action Agreed	Resolved	Authorised human	Immediate completed outcome with category evidence/approval.	RESOLVED.
Monitoring	Resolved	Authorised human	Monitoring result, category evidence, residual risk and approvals.	RESOLVED.
Monitoring	Investigating	Owner/manager after trigger	Commitment breach, reversal, failed action or new cause; reason/evidence.	REOPENED or MONITORING_FAILED; no SLA reset.
Resolved	Closed	Authorised closure reviewer	Checklist complete; monitoring satisfied; evidence resolvable; approvals valid; reviewer separation where required.	CLOSED; final SLA/outcome/limitations.
Resolved	Investigating	Human after system/human detection	Resolution evidence invalidated; manager approval if accepted residual risk was involved.	RESOLUTION_WITHDRAWN and REOPENED; no reset.
Suppressed	Open	System	Expiry/revocation/material reactivation; no valid eligible owner.	SUPPRESSION_ENDED and REOPENED; remaining SLA resumes.
Suppressed	Assigned	System	Expiry/revocation/material reactivation; valid owner exists.	SUPPRESSION_ENDED and REOPENED; remaining SLA resumes.
Closed	New linked Open episode	System	Material post-closure recurrence under frozen rule.	SUCCESSOR_OPENED and REOPENED link; predecessor unchanged.

Same-state controlled actions

- Risk refresh, priority/criticality override, reassignment/delegation, escalation, reminder, SLA breach, evidence/note addition, approval request/decision, possible-resolution detection and audit annotation do not create lifecycle states.
- Every owner/priority/SLA/evidence change appends before/after and authority context. Calculated score/contribution history is immutable; an approved operational priority is an overlay with expiry.
- A rejected approval applies no requested effect. An approved request grants authority but does not alter state/value until a separate controlled execution event occurs.

7. Validation message catalogue

ID	Condition	User message	UI response
VAL-001	Invalid transition	This action is not available from {state}. No changes were saved.	Keep user on detail; show eligible next actions.
VAL-002	Unauthorised role	Your simulated role is not authorised for this action. No changes were saved.	Do not reveal restricted content; offer read-only return.
VAL-003	Wrong owner	Only the current owner or an authorised manager can begin this investigation.	Offer reassignment request if permitted.
VAL-004	Missing owner	Select one active, eligible owner before assigning this exception.	Focus owner field.
VAL-005	Critical reassignment	Critical reassignment requires an approved, unexpired authority reference.	Link to request approval; no owner change.
VAL-006	Self-approval	You cannot approve this material request because you created it. Select a distinct authorised approver.	Block decision; record no approval.
VAL-007	Evidence required	Add a resolvable evidence reference for {action/category}.	Focus evidence section; retain draft only in session.
VAL-008	Broken evidence	Evidence reference {id} cannot be verified. Replace it with a valid synthetic reference or use the exceptional route.	Block normal resolution/closure.
VAL-009	Suppression expiry	Suppression requires a future expiry within the permitted term: Critical 1 business day; otherwise 5 business days.	Focus expiry; no request/event.
VAL-010	Suppression overlap	An effective or pending suppression already covers this episode and period.	Open existing request/record.
VAL-011	Monitoring condition	Define a measurable condition and expected event or date before Monitoring can begin.	Focus condition/date.

ID	Condition	User message	UI response
VAL-012	Unresolved action	Resolution is blocked while mandatory action {id} remains open or overdue.	Link to action.
VAL-013	Residual risk approval	Material residual risk requires an approved, unexpired acceptance with required concurrence.	Link to approval request.
VAL-014	Direct closure	Only a Resolved exception can be Closed. Investigate and resolve with evidence first.	No closure form rendered for other states.
VAL-015	Closure checklist	Closure is incomplete: {missing checklist items}.	Keep form; focus first missing item.
VAL-016	Possible resolution	This evidence does not change lifecycle state. Review it and submit a human resolution if appropriate.	Information message, not error.
VAL-017	Score language	Enter a risk-priority score between 0 and 100; do not enter or label it as a probability.	Applies only to controlled test/config display; users cannot edit calculated score.
VAL-018	Missing signal	{signal} is unavailable or invalid and cannot be treated as zero. Review source status.	Show affected rules and freshness.
VAL-019	Date chronology	{field} must be after {comparison field} and within the approved control window.	Focus field; show Melbourne time.
VAL-020	Quantity/UOM	Quantity must be greater than zero and use the line's controlled base UOM or a valid conversion.	Block evidence/action calculation.
VAL-021	Duplicate submit	This action was already recorded as event {event_id}. No duplicate event was created.	Show existing event.
VAL-022	Concurrent change	This exception changed after you opened it. No changes were saved. Reload and review the new state, owner and approvals.	Discard server submit; preserve draft text for copy.
VAL-023	Stale publication	Operational data is {age} old, above the 26-hour freshness threshold. Workflow history may be newer; verify source facts before acting.	Persistent banner; do not silently disable safe read-only review.
VAL-024	Failed latest attempt	The latest run attempt failed and was not published. Current screens remain on successful run {run_id}.	Link Pipeline Health.
VAL-025	Historical immutability	Historical events cannot be edited or deleted. Add a linked correction or annotation instead.	Offer permitted append action.
VAL-026	Approval conditions	Approval conditions are not yet satisfied: {conditions}. The requested change cannot be applied.	Link condition evidence.

ID	Condition	User message	UI response
VAL-027	Approval expired	Approval {id} expired at {time}. Submit a new request; the prior decision remains in history.	No execution.
VAL-028	No results	No exceptions match these filters. Clear filters or return to your default work view.	Not an error; show clear-filter action.
VAL-029	Run not publishable	Publication is blocked because critical validation or reconciliation checks failed.	No current pointer change.
VAL-030	Export scope	Choose a bounded export scope and confirm it contains synthetic data only.	No unbounded all-history export.

8. Confirmation and approval flows

Flow	Applies to	User/system sequence	Control outcome
Standard append	Add note/evidence/checkpoint	Review fields -> Save entry	No destructive confirmation; append event and show receipt ID/time. Duplicate correlation returns existing event.
State transition	Begin investigation; agree action; begin monitoring; normal resolution	Complete form -> Review summary -> Confirm transition	Recheck role/state/version/fields; commit domain record and transition event atomically.
Assignment	Assign/reassign	Select eligible owner -> reason/handover -> authority check -> Confirm	State changes Open->Assigned only on first valid owner. Later reassignment retains state and SLA. Critical route requires approval.
Material request	Priority reduction, suppression, criticality override, residual-risk acceptance, exceptional closure, SLA extension, due-date concession, critical reassignment	Draft request -> validate evidence/expiry -> submit -> pending	No requested effect. REQUESTED event records original/requested values and required roles.
Approval decision	Approve / conditional / reject	Open request -> review evidence/authority/SoD -> rationale -> confirm	Decision event only. Rejection has zero effect. Conditional approval cannot execute until conditions are evidenced.
Apply approved control	Execute suppression/override/extension/reassignment/exceptional closure	Open approved request -> verify validity/conditions -> confirm Apply	Separate APPLY event; approval remains linked. Approve-and-apply convenience is permitted only when UI creates two ordered events and approver is also authorised executor.
Resolution	Human enters Resolved	Choose category -> conditional fields -> evidence/residual risk -> approvals -> review -> confirm	No automatic path. Current score and possible-resolution evidence are context only.
Closure	Human enters Closed	Resolved-only checklist -> evidence monitoring/SLA -> reviewer separation -> confirm final closure	CLOSED event atomically freezes current state projection. Exceptional path requires manager plus governance concurrence.
Suppression expiry/revocation	Return to Open/Assigned	System at expiry or human revocation -> eligibility/owner check	SUPPRESSION_ENDED and REOPENED events; prior stage retained for audit; SLA resumes remaining time; no reset.
Receipt correction/recurrence	Invalidate evidence	System detects -> possible/reversal event -> human review if pre-close; system successor if post-close threshold met	Before Closed, same episode returns to Investigating only through controlled human transition. After Closed, predecessor stays Closed and one linked successor opens.

Approval-specific rules

Request type	Primary approver	Additional concurrence	Term / constraint
Risk-priority reduction	Supply-chain manager	Governance if Critical becomes Medium/Monitor	Time-bound; calculated score and prior SLA remain.
Suppression	Supply-chain manager	Governance for Critical >1 business day, repeat or second extension	Critical max initial 1 business day; otherwise 5; evidence/expiry mandatory.
Product-criticality override	Supply-chain manager	Governance for downward Critical change	Effective-dated; historical scores unchanged.
Residual-risk acceptance	Supply-chain manager	Governance for Critical or safety/production-critical impact	Expiry/review required; recurrence remains active.
Exceptional closure	Supply-chain manager	Governance/audit mandatory	Explicit limitation; alternative corroboration; not convenience.
Critical reassignment	Supply-chain manager	Second manager/governance if manager requested change	No SLA reset; handover required.
SLA extension	Supply-chain manager	Governance for Critical or repeat	Prospective; once only; never erases breach.
Due-date extension/concession	Supply-chain manager	Planner/demand owner; governance for Critical	Original date retained; distinguish correction from concession.

9. Empty, loading, failure and stale-data states

Context	Required presentation and behaviour
Control Tower - no active exceptions	Show 'No active exceptions in the latest successful publication' plus publication/run/as-of. Do not imply the system is risk-free; provide Queue and Pipeline Health links.
Queue - filters return zero	Show applied filters and Clear filters. Preserve query; do not replace with unfiltered data automatically.
Queue - no owner records	Show UNASSIGNED and escalation/SLA; never fabricate generic owner.
Approvals - empty	Show 'No requests require this identity's decision' and links to all/previous decisions if authorised.
Receipts/schedules - none	Show 'No effective receipt/schedule events available as of {time}' with source freshness and missing/invalid status, not an empty grid.

Context	Required presentation and behaviour
Audit - no events	For a valid Open episode this is a control defect. Show blocking error, reference ID and Pipeline/Audit link; do not allow writes.
Initial page load	Skeleton for header/cards/table with aria-busy. Keep navigation usable; announce completion politely; no fake zero values.
Long filter/export	Show progress label with operation and scope; allow safe cancel before server mutation; do not block the entire navigation if avoidable.
Section-only failure	Keep successfully loaded episode header and other tabs; mark failed section with retry/reference. Disable actions whose validation depends on missing section.
Whole-record failure/not found	Do not show partial sensitive context. Provide reference ID and safe return to preserved queue.
Latest run failed	Continue to serve last successful publication; red text/icon banner states failed attempt ID/time and 'not published'. Pipeline Health provides defect detail.
Published data stale	At >26 hours, show persistent stale banner, exact age and affected sources. Permit read/append only where safe policy allows; require explicit acknowledgement on evidence-based decisions.
Source-specific stale/missing	Show per-source age/status and affected rules. Score remains Qualified/Invalid according to contract; no blank or zero substitution.
Record changed while form open	Reject stale submit with zero mutation; show changed fields/state/owner/approval summary; preserve user draft for copy; require reload.
Duplicate submit/retry	Return the prior successful event using correlation key; do not create a second note, transition or decision.
Approval expired while open	Reject execution; show expiry and create no effect. Allow new request prefilled from prior values but not copying prior approval authority.
Evidence link unavailable	Mark Broken/Unavailable, show last verified time and block routes requiring it. Offer append replacement/correction; do not delete original.
Pipeline reconciliation failure	Status FAILED/NOT PUBLISHED; show source, accepted, quarantined, rejected and unexplained difference. No current dataset pointer change.

10. Accessibility and usability requirements

Area	Requirement
Target	Aim for WCAG 2.2 AA for the MVP interface. Document Streamlit/component exceptions rather than claiming blanket compliance.
Keyboard	All navigation, filters, tabs, pagination, forms, dialogs and row drill-through operable without a mouse; logical focus order; visible focus indicator; no keyboard trap; Escape closes non-destructive dialogs.
Screen reader	Semantic headings/landmarks; every input has persistent label/instruction; table headers announced; icon-only controls have accessible names; state/priority/SLA changes announced through polite live regions.
Colour/contrast	Text contrast $\geq 4.5:1$; large text and component boundaries $\geq 3:1$. Priority, SLA, missing data, approval and run status always include text/shape/icon in addition to colour.
Errors	On submit, show an error summary with links to fields; inline message describes cause and fix; focus moves to summary/first error; valid data is retained.
Zoom/reflow	Core tasks remain usable at 200% zoom and at 1024px desktop width. Wide tables may horizontally scroll with sticky identifiers/headers; detail converts to stacked sections. Mobile write workflows are not an MVP claim.
Targets	Interactive targets at least 24 by 24 CSS pixels or sufficient spacing; primary action has clear label; destructive-equivalent decisions are separated from navigation.
Language	Use operational verbs: Assign, Begin investigation, Agree action, Begin monitoring, Resolve, Close, Request suppression. Never use Candidate Risk, Escalated, Breached or Reopened as state labels.
Dates/numbers	Display '20 Jul 2026, 14:05 AEST/AEDT' with Australia/Melbourne label; ISO value available in detail/export. Show quantity plus UOM and currency. Never encode unknown as zero.
Cognitive load	Default Queue shows operational essentials; detailed source/history remains on tabs. Progressive disclosure reveals conditional fields by action category. Confirmation summaries use before/after values.
Context preservation	Back to queue restores filters/sort/page. Draft form warns before navigation. Successful action returns to detail and highlights the new immutable event.
Performance feedback	Default 1,500-line queue target: 95th percentile ≤ 3 seconds across the defined test protocol. Use pagination; never render all rows to imply speed.
Help	Inline definitions for score, effective priority, state, evidence, business-time SLA, suppression, possible resolution and simulated authentication. Provide a keyboard-accessible control legend.

11. Audit evidence generated by each action

Action	Event type	Minimum reconstruction payload
Open episode	EPISODE_OPENED	Episode/line/site; candidate/run/rule/score; opening reason; source snapshots; actor System; SLA obligations; dedup result.
Automatic assignment	OWNER_ASSIGNED	Owner/queue; mapping route/version; eligibility; assigned-at; System actor; prior/new; assignment SLA outcome.
Manual assign/reassign	OWNER_CHANGED	Prior/new/base/delegate owner; actor/role; reason; handover; approval; effective time; state unchanged except Open->Assigned; SLA no-reset.
Begin investigation	INVESTIGATION_STARTED	Prior/new state; investigator; validity assessment; cause/status; reviewed facts; next review; evidence; residual exposure.
Add note/contact	NOTE_ADDED / CONTACT_RECORDED	Note/contact type; text; channel/party synthetic reference; actor/time; related action/evidence; current state.
Add/correct evidence	EVIDENCE_ADDED / EVIDENCE_CORRECTED	Evidence metadata/purpose/source/time/integrity; original link if corrected; actor; no binary content copied into audit.
Agree action	ACTION_AGREED	Category; rationale; action owner; target/commitment; next action/date; evidence; residual exposure; approval; monitoring expectation; state transition.
Amend/withdraw plan	ACTION_AMENDED / ACTION_WITHDRAWN	Prior action reference; changed fields; reason/new facts; actor/time; approvals; prior plan retained; optional transition to Investigating.
Begin monitoring	MONITORING_STARTED	Initiation evidence; condition/metric; expected event/date; next review; reminders; owner; residual exposure; state transition.
Monitoring checkpoint	MONITORING_CHECKPOINT	Observed value/status; evidence; commitment result; next review; actor/time; possible breach linkage.
Risk refresh	RISK_RECALCULATED	Candidate evaluation; score/band; full contribution set; missing signals; run/rule/as-of/source freshness; state unchanged.
SLA reminder/breach/escalation	SLA_REMINDER / SLA_BREACHED / ESCALATED	Obligation; due/breach time; elapsed/paused minutes; calendar version; severity; owner/manager accountability; source event; state unchanged.
SLA pause/resume/extension	SLA_PAUSED / SLA_RESUMED / SLA_EXTENDED	Reason; interval; consumed/remaining; original/revised due; approval/conditions; actor; prior breach preserved.
Approval request	APPROVAL_REQUESTED	Type; original/requested values; requester/role; reason/evidence; required approvers/concurrence; expiry; conditions; target action.
Approval decision	APPROVAL_APPROVED / CONDITIONAL / REJECTED	Decision-maker/role; independence result; rationale; conditions/expiry; evidence reviewed; no requested effect.

Action	Event type	Minimum reconstruction payload
Apply approved control	APPROVED_CHANGE_APPLIED	Approval reference; executor; before/after; effective/expiry; condition evidence; separate target-specific event.
Suppression	SUPPRESSION_APPLIED	Prior state/owner; reason/scope; requester/approver; start/expiry/review; evidence; conditions; score context; SLA consumed/remaining; recurrence criteria.
Suppression end	SUPPRESSION_EXPIRED / REVOKED plus REOPENED	Suppression reference; trigger; prior/target state; eligible owner; latest candidate; SLA resumed/no reset; actor System or authorised human.
Possible resolution	POSSIBLE_RESOLUTION_DETECTED	Source event; proposed category; quantity/date/source; confidence/missing warnings; related receipt/correction; review status; no state change.
Review possible resolution	POSSIBLE_RESOLUTION_REVIEWED	Reviewer; accept-for-form/dismiss/investigate; rationale; linked resolution or correction; no automatic state change.
Resolve	RESOLVED	Category/cause; resolver/time; evidence; outcome quantity/date; monitoring result; residual-risk statement; approval; closure-review date; current score/rule; prior/new state.
Withdraw resolution	RESOLUTION_WITHDRAWN plus REOPENED	Invalidating observation; prior resolution; evidence/correction; actor; approval when required; target Investigating; SLA no-reset.
Close	CLOSED	Resolution reference; reviewer/time; checklist; closure rationale; evidence/approvals; final residual risk/outcome; SLA result; limitations/exceptional flag.
Post-close recurrence	SUCCESSOR_OPENED plus REOPENED	Predecessor/successor; new candidate/run/rule; recurrence reason/evidence; new SLA obligations; predecessor remains Closed.
Data correction/reprocess	DATA_CORRECTION_RECORDED / RUN_AUTHORISED	Admin; reason; affected objects; before/after/checksum; source authority; corrected run; publication outcome; prior run unchanged.
Audit annotation/export	AUDIT_ANNOTATION_ADDED / AUDIT_EXPORT_GENERATED	Reviewer; observation/scope; filters; event range/count; publication/version; generated-at; file hash/reference.

HISTORICAL CORRECTION RULE

Normal UI roles cannot edit or delete an event. If a note, evidence reference, receipt allocation, approval or source fact is wrong, the user appends a correction/reversal that identifies the original event, reason, actor, before/after and downstream impact. Current projections may change; history does not.

12. Given/When/Then acceptance criteria

ID	Acceptance criterion
AC-UI-001	Given any page is opened, when the header renders, then SYNTHETIC DATA, the named simulated identity/role, non-production-authentication limitation, latest successful run and Melbourne time context are visible.
AC-UI-002	Given a deep link, when a user lacks page/action permission, then restricted content is not rendered and a safe unauthorised page is shown.
AC-Q-001	Given the 25-record ranking fixture, when the Queue opens with control ranking, then rows follow the frozen tier, severity, date, criticality, inactivity, opened-at and exception-ID precedence exactly.
AC-Q-002	Given each required filter and every approved two-filter fixture, when filters are applied, then the exact matching set and active filter chips are shown.
AC-Q-003	Given 184 results and page size 50, when page 2 opens, then rows 51-100 under the stable sort are shown and filters remain unchanged.
AC-Q-004	Given a score-qualified exception, when its row renders, then unavailable/invalid signals are labelled and none is displayed as zero or blank.
AC-Q-005	Given an approved operational priority reduction, when the row renders, then the effective band, calculated band/score, override marker and expiry are all discoverable.
AC-Q-006	Given the user drills into an episode and returns, when Back to queue is selected, then the prior filters, sort, page and result position are restored.
AC-D-001	Given a candidate snapshot, when Risk & Source opens, then every configured component including unavailable components, observed value, threshold, gross/applied points, cap, total, rule version, run and as-of time are shown.
AC-D-002	Given a hand-calculated fixture, when the score detail renders, then stored contributions reproduce the 0-100 score and exact band without probability language.
AC-D-003	Given split schedules and partial receipts, when Receipts opens, then schedules remain within one episode and residual quantity reconciles to ordered quantity, allocations, corrections and approved cancellations.
AC-D-004	Given a receipt correction, when history renders, then the original receipt remains, the correction links to it and no cumulative receipt event is overwritten.
AC-D-005	Given possible-resolution evidence, when Summary opens, then the evidence is visible but current state remains unchanged until an authorised human submits Resolution.

ID	Acceptance criterion
AC-W-001	Given Open with a valid mapping, when assignment executes, then one owner is recorded, Open becomes Assigned, prior/mapping context is audited and the assignment clock stops.
AC-W-002	Given Open without a valid mapping, when opening completes, then the episode stays Open/UNASSIGNED, remains visible and the correct reminder/breach events occur.
AC-W-003	Given Assigned, when the owner submits complete investigation-start fields, then state becomes Investigating and actor/time/evidence are appended.
AC-W-004	Given Investigating with an incomplete plan, when Action Agreed is attempted, then the transition is blocked, exact fields are identified and no domain event/state change occurs.
AC-W-005	Given Action Agreed, when initiation evidence and measurable condition/date are submitted, then state becomes Monitoring and a monitoring obligation is created.
AC-W-006	Given Monitoring and a missed commitment, when the breach is detected, then breach/escalation events append; the owner remains; state changes only through the authorised return-to-Investigating action.
AC-W-007	Given a lower refreshed score or zero residual quantity, when scoring publishes, then a score/possible-resolution event appends and Resolved/Closed do not change automatically.
AC-W-008	Given an allowed resolution category, when an authorised human submits complete evidence/residual-risk/approval fields, then state becomes Resolved and a closure-review obligation begins.
AC-W-009	Given any state other than Resolved, when closure is attempted through UI or direct request, then it is blocked and no state mutation occurs.
AC-W-010	Given Resolved and a complete normal checklist, when an authorised reviewer confirms closure, then state becomes Closed and final evidence/SLA/limitations are appended.
AC-W-011	Given exceptional closure, when either manager or governance concurrence is missing, then Closed is blocked; with both distinct decisions and complete limitation evidence, closure succeeds.
AC-A-001	Given a material request, when its requester attempts approval under any switched role, then self-approval is blocked using identity, not role label, and no approval decision is created.
AC-A-002	Given an authorised distinct approver rejects a request, when the decision is confirmed, then rejection/rationale append and the requested state/value has zero effect.
AC-A-003	Given an approved-with-conditions request, when conditions lack evidence, then Apply is blocked; after condition evidence is appended, a separate Apply action is available.
AC-S-001	Given an active episode, when suppression is requested without reason, evidence or expiry, then submission is blocked and no request is created.

ID	Acceptance criterion
AC-S-002	Given an approved suppression, when it is applied, then prior state/owner and SLA consumed/remaining are retained, state becomes Suppressed and normal queue views exclude it while suppression views include it.
AC-S-003	Given suppression expires and an eligible owner exists, when expiry processing occurs, then state becomes Assigned, REOPENED/SUPPRESSION_ENDED append and SLA resumes without reset.
AC-R-001	Given resolution evidence is reversed before closure, when an authorised human accepts invalidation, then the same episode returns to Investigating with linked Reopened/Resolution Withdrawn events and no SLA reset.
AC-R-002	Given material recurrence after Closed, when the frozen recurrence rule triggers, then exactly one linked successor opens with new SLA clocks and the predecessor remains Closed.
AC-SLA-001	Given Melbourne after-hours, weekend, holiday and DST fixtures, when due times are calculated, then business minutes match the frozen calendar and UTC offsets are not hard-coded.
AC-SLA-002	Given reassignment or severity decrease, when the change is applied, then no clock resets or extends without separate prospective approval and prior breach remains visible.
AC-SLA-003	Given supplier waiting, when a pause is requested, then it is blocked; given an approved data/system hold or effective suppression, consumption pauses and later resumes.
AC-P-001	Given a successful publication followed by a failed run, when any page opens, then the successful run remains current and the failed attempt is clearly labelled NOT PUBLISHED.
AC-P-002	Given failed critical validation or reconciliation, when publication is attempted, then no current pointer changes and Pipeline Health shows source/accepted/quarantined/rejected totals and the unexplained difference.
AC-P-003	Given published data older than 26 hours, when operational pages open, then a persistent stale banner shows exact age, affected sources and safe next action.
AC-AUD-001	Given a golden episode, when Audit renders/exports, then event order and every mandatory actor, time, before/after, reason, evidence/approval, run/rule and correlation field match the expected ledger.
AC-AUD-002	Given any standard role, when edit/delete of historical history is attempted, then it is unavailable or blocked; a linked correction path remains available where authorised.
AC-CON-001	Given two sessions on the same episode, when the second submits against an old record version, then the action is rejected with zero mutation and the draft remains available for copy.
AC-ACC-001	Given keyboard-only use, when a user completes Queue-to-action-to-history, then every control is reachable in logical order, focus is visible and no keyboard trap occurs.

ID	Acceptance criterion
AC-ACC-002	Given status indicators are viewed without colour, when priority/SLA/missing/approval/run states render, then text/icon/shape still communicates each state.
AC-PERF-001	Given the normal 1,500-line open workload and defined environment, when the default queue is requested 30 times, then 95th percentile response is <=3 seconds or the miss is reported honestly.

13. MVP versus deferred UI features

Capability	First Streamlit release	Deferred / excluded
Identity/access	Named simulated identities/roles; role-based action gating; persistent limitation label; direct-link re-authorisation.	Production SSO/MFA, identity lifecycle, enterprise RBAC/SoD, penetration testing and admin-resistant security.
Operational pages	Control Tower, Queue, Detail, Approvals, Pipeline Health, Audit Review and Help.	Mobile-first application, bespoke workflow designer, user-configurable dashboards and enterprise case portal.
Queue	Documented sort, filters, pagination, indicators and drill-through; no bulk mutation.	Saved/shared views, bulk assignment/action/suppression, advanced personalisation and real-time collaborative refresh.
Workflow	Eight states, approved transitions, assignment, notes, action, monitoring, resolution, closure, suppression and reopen/successor demonstrations.	Custom state machine, autonomous resolution/closure and cross-order orchestration.
Approvals	One generic request/decision structure; required role/identity checks; conditions, expiry and separate apply.	Enterprise delegation-of-authority engine, digital signatures and external approval routing.
Evidence	Synthetic metadata/reference, purpose, source, integrity status and resolvability checks.	Binary document repository, confidential attachments, OCR and external content management integration.
Pipeline	Run/step/freshness/defect/quarantine/reconciliation/publication health; attributable manual corrected-reprocess authorisation.	GitHub Actions scheduling, 24/7 monitoring/on-call, automatic recovery and production observability.
Audit	Append-only event pattern, cross-episode read-only search/export, correction linkage and completeness view.	WORM/tamper-proof storage, legal retention, backup/restore certification and database-admin separation proof.
Integration	Synthetic source snapshots and references only.	Live ERP/EDI/WMS/supplier portal, automatic supplier contact, order change or logistics execution.
Reporting	Operational counts and watchlists only; curated publication identifiers for later BI reconciliation.	Power BI measure/UI implementation inside Streamlit; enterprise Power BI service/gateway.

Capability	First Streamlit release	Deferred / excluded
Notifications/AI	None in first release; in-app queues and banners only.	Email/Teams/Slack; AI summaries/options/drafts; any autonomous action.

14. Critical self-review

Review area	Weakness / risk	Required treatment
Queue density	The requested queue fields can create a wide, cognitively expensive grid.	Keep the frozen default operational columns, use sticky identifiers and progressive disclosure; test at 1024px/200% zoom. Do not solve it with unreadably small text.
Streamlit security	Hidden buttons and session state are not security boundaries; reruns can create stale or repeated submissions.	Recheck role/state/version on every submit, use idempotent correlation, and state clearly that authentication/SoD are simulated.
Human control workload	Forcing human Resolved and Closed increases effort, especially when receipts are objective.	Retain as an intentional MVP safeguard. Measure review workload before proposing controlled automation in a later phase.
Approval complexity	Eight material routes, concurrence and separate execution could over-engineer a solo MVP.	Use one generic approval object/form with configuration-driven requirements; implement only frozen routes and fixtures, not a workflow engine.
Suppression risk	Suppression can hide urgent work and blur the difference between candidate blocking and an operational state.	Only active episodes can enter Suppressed; every suppression is approved/evidenced/time-bounded; default management views show suppressed/expiring counts.
Operational override	Showing effective priority may visually demote a calculated Critical case.	Always retain calculated score/band and override badge; provide calculated-Critical filter and expiry; never overwrite contributions/SLA history.
Stale data	Allowing workflow writes against stale source facts can create bad decisions; blocking all writes can also halt legitimate case management.	Display source-specific staleness and require acknowledgement for evidence-dependent transitions; block only actions whose required evidence is unavailable/invalid. Freeze exact policy in tests.
Control Tower overlap	Operational cards may drift into a second management dashboard and conflict with Power BI.	Limit to current workload, approvals, SLA and data confidence with drill-through. Trends, outcome quality and management analytics remain in Power BI.
Timeline scale	Append-only score, receipt and event histories can become long and slow.	Default to grouped chronological summaries with filters and server pagination; event detail remains available. Never truncate exports silently.
Evidence strength	A resolvable synthetic reference does not prove that evidence is authentic or legally retained.	Label the limitation; test linkage/completeness only; do not claim document-management or tamper-proof assurance.
Role ambiguity	A manager can also be operational owner, making reviewer independence unclear.	Evaluate identity-level conflicts per request and category. Require governance concurrence where frozen; disclose that simulated identities cannot prove enterprise independence.

Review area	Weakness / risk	Required treatment
SLA precision	Proposed Melbourne hours, holidays and targets are assumptions, not validated policy.	Label them as proposed portfolio assumptions, version the calendar, test DST/holiday boundaries and avoid claims of organisational acceptance.
Possible-resolution pressure	A prominent flag may anchor users toward premature resolution.	Use neutral wording 'Evidence available for review', show reversal/missing warnings, and offer Dismiss/Investigate before Resolve.
Pipeline truth	Users may confuse latest attempt with latest successful publication.	Show both in every relevant header; failed attempts are never labelled current and never advance the publication pointer.
Concurrency	Streamlit's rerun model and a solo demo cannot prove robust multi-user transactions.	Use record-version conflict checks and atomic event/state write expectations; disclose tested writer limit and defer enterprise-scale concurrency.
Accessibility	Third-party table components may not meet keyboard/screen-reader expectations.	Use native accessible components where possible, test the exact component/version, publish exceptions and provide a card/list fallback if necessary.

OVERALL ASSESSMENT

The specification is credible for a portfolio-grade control demonstration if it is implemented as a thin, deterministic interaction layer over the frozen state/event contracts. The main credibility risk is not missing features; it is implying production security, immutability, real organisational validation or business benefit that the Streamlit/synthetic MVP cannot prove.

Appendix A. Final page map

Page	Children / tabs	Purpose
App shell	Role switch; Help/Control Legend	Global simulation, publication, freshness and time-zone context.
Control Tower	Filtered KPI/watchlist drill-through	Current operational control summary.
Exception Queue	Queue table -> Exception Detail	Ranked work and historical inclusion.
Exception Detail	Summary Risk & Source Receipts Work Approvals SLA Audit	One episode workspace; action rail is state/role-aware.
My Approvals	Approval list -> Approval detail -> decision -> separate Apply	Material authority and concurrence.
Pipeline Health	Publication -> run attempt -> step -> defect/quarantine/control totals	Data/run confidence and attributable reprocessing.
Audit Review	Search -> event detail -> bounded export	Cross-episode assurance and evidence.
Safe failure	Unauthorised not found stale conflict unavailable section	No leakage, no partial mutation and stable return path.

Page-to-page movement

- Control Tower cards open the Queue with visible fixed filter chips; watchlist items may open Detail directly while preserving a one-row result context.
- Queue opens Detail in the same application context. Detail exposes Previous/Next only within the frozen current result set and returns to the exact Queue view.
- Detail can open a linked Approval or Pipeline run in context; those pages return to the same exception/tab.
- Audit event detail may link to its exception, candidate/run/rule, evidence and approval reference subject to the user's read permission.
- No page transition is itself a business event. Only confirmed controlled actions create immutable events.

Appendix B. Complete interaction matrix

ID	Screen	Interaction	Role	Precondition	Result / event
I01	Shell	Switch simulated identity/role	Any	Named configured pairing	Session context only; no business event
I02	Tower	Open filtered queue	View roles	Fixed filter set visible	No mutation
I03	Queue	Search/filter/sort/page	View roles	Valid values; stable tie-breaker	No mutation; optional usage telemetry deferred

ID	Screen	Interaction	Role	Precondition	Result / event
I04	Queue	Open detail	View roles	Authorised episode	No mutation; preserve queue context
I05	Detail	Select candidate snapshot	View roles	Snapshot belongs to episode	No mutation
I06	Detail	Assign	Analyst/manager/System	Open; eligible owner; reason/approval if manual Critical	Open->Assigned; OWNER_ASSIGNED
I07	Detail	Reassign/delegate	Analyst/manager within authority	Active; eligible owner; reason/handover; approval if Critical	State same; OWNER_CHANGED; SLA no reset
I08	Detail	Begin investigation	Owner/manager	Assigned; complete start fields	Assigned->Investigating
I09	Work	Add note/contact	Analyst/planner/manager	Active/Resolved as permitted; note type/text	Same state; append event
I10	Work	Add/correct evidence	Operational/admin/governance by purpose	Metadata complete; correction links original	Same state; append event
I11	Work	Agree action	Owner/planner/manager	Investigating; full plan/approval	Investigating->Action Agreed
I12	Work	Amend/withdraw action	Owner/manager	Action Agreed; reason/new evidence	Same state or ->Investigating; prior retained
I13	Work	Begin monitoring	Action owner	Action Agreed; initiation evidence/condition/date	Action Agreed->Monitoring
I14	Work	Record checkpoint	Owner/planner/manager	Monitoring; observed result/evidence/next review	Same state; MONITORING_CHECKPOINT
I15	Work	Return to investigation	Owner/manager	Action Agreed/Monitoring; plan failure/breach/new cause	->Investigating; no SLA reset
I16	Work	Review possible resolution	Authorised resolver	Flag belongs to current episode	Dismiss/investigate/open form; state unchanged
I17	Work	Submit normal resolution	Authorised analyst/planner/manager	Investigating/Action Agreed/Monitoring; category evidence; no open mandatory action	->Resolved
I18	Work	Submit residual-risk resolution	Manager	Approved unexpired acceptance; concurrence if required	->Resolved
I19	Detail	Withdraw resolution	Resolver/manager	Resolved; invalidating evidence; approval if prior accepted risk	Resolved->Investigating; REOPENED
I20	Detail	Close normal	Manager/designated reviewer	Resolved; checklist/monitoring/evidence; independence	Resolved->Closed
I21	Detail	Close exceptional	Manager executor	Resolved; manager approval + governance concurrence + limitation	Resolved->Closed; exceptional flag

ID	Screen	Interaction	Role	Precondition	Result / event
I22	Detail	Request suppression	Operational owner/manager	Open-Monitoring; reason/evidence/expiry/scope	Pending request; no state change
I23	Approvals	Decide suppression	Distinct manager/governance	Authority/SoD/evidence/term	Decision only; no state change
I24	Detail/Approval	Apply suppression	Authorised executor	Approved valid request; conditions satisfied	Active->Suppressed; SLA pause
I25	Detail	Revoke suppression	Manager	Suppressed; reason	System/transaction ->Open/Assigned; SLA resume
I26	System	Expire/reactivate suppression	System	Expiry/material trigger	->Open/Assigned; REOPENED; no reset
I27	Approvals	Request material override	Authorised operational user	Original/requested/reason/evidence/ expiry	APPROVAL_REQUESTED; no effect
I28	Approvals	Approve/conditional/reject	Distinct required approver(s)	Authority, SoD, evidence, rationale	Decision event; reject zero effect
I29	Detail/Approval	Apply approved override	Authorised executor	Unexpired approval and conditions	Overlay/extension/change event; score unchanged
I30	System	Refresh score	System	Successful published run	RISK_RECALCULATED; state unchanged
I31	System	Record SLA reminder/breach/escalation	System	Clock condition under calendar version	Same state/owner; accountability event
I32	System	Detect possible resolution	System	Objective source event	POSSIBLE_RESOLUTION_DETECTED only
I33	System	Open linked successor	System	Closed predecessor; material recurrence; dedup passes	New Open episode; predecessor unchanged
I34	Pipeline	View run/defect	Admin/manager/audit/reviewer	Read permission	No mutation
I35	Pipeline	Record correction/authorise reprocess	Admin; manager accountable	Reason, affected objects, source authority, before/after	Admin audit event; new attempt; old attempt unchanged
I36	Audit	Add annotation	Governance reviewer	Observation and event/episode scope	AUDIT_ANNOTATION_ADDED
I37	Audit	Generate bounded export	Authorised view roles	Filters/scope/synthetic confirmation	AUDIT_EXPORT_GENERATED

Appendix C. Form-field and validation catalogue

Form	Fields (* required)	Validation	Audit output
F01 Assignment	New owner*; assignment basis*; reason* if manual; handover; effective time (server); approval ref if Critical reassignment; record version*.	Owner active/eligible; one current owner; no self-approved Critical change; no SLA reset; optimistic version current.	OWNER_ASSIGNED/OWNER_CHANGED.
F02 Investigation start	Validity assessment*; cause/status*; source facts reviewed*; investigator (current)*; supplier/planner contact status; next review*; evidence refs; residual exposure acknowledgement*; note.	Assigned only; owner/manager; next review future; invalid/missing facts explicit.	INVESTIGATION_STARTED.
F03 Note/contact	Type*; note/contact summary*; synthetic party/channel; occurred-at; evidence refs; related action; record version*.	No blank text; no real personal data; occurred-at not future; historical note correction uses new form.	NOTE_ADDED/CONTACT_RECORDED.
F04 Evidence	Evidence type*; synthetic reference*; purpose*; source/creator*; effective/recorded time*; item/site/quantity/date coverage where applicable; integrity/resolvable status*; note.	Reference unique/valid format; required domain linkage; no raw confidential document; corrections link original.	EVIDENCE_ADDED/EVIDENCE_CORRECTED.
F05 Action agreement	Action category*; rationale*; action owner*; agreed-at (server)*; target/commitment*; quantity/UOM/date conditional; next action and due*; evidence*; residual exposure*; approval refs conditional; monitoring condition/event/date*; record version*.	Investigating only; complete conditional fields; eligible owner; future dates; UOM valid; approvals complete; condition measurable.	ACTION_AGREED and state transition.
F06 Monitoring start	Initiation time*; initiation evidence*; condition/metric*; expected event/date*; expected quantity/UOM conditional; next review*; owner*; reminder rule; residual exposure*; record version*.	Action Agreed only; condition measurable; expected/next dates valid; approval remains effective.	MONITORING_STARTED and obligation.
F07 Monitoring checkpoint	Checkpoint time*; condition result*; observed value/unit; status*; evidence* when met/failed; next review conditional; breach/new-cause reason conditional; record version*.	Monitoring only; values compatible; failed condition requires next action or return decision; cannot silently resolve.	MONITORING_CHECKPOINT; optional breach event.
F08 Resolution	Category*; cause*; resolver/time*; category evidence*; outcome quantity/date conditional; monitoring result*; residual-risk statement/amount*; approval refs conditional; closure-review date*; current score/rule ref*; limitations; record version*.	Allowed source states only; authorised resolver; no open mandatory action; category evidence/period complete; residual approvals valid; human confirm.	RESOLVED and closure-review obligation.
F09 Closure	Resolution ref*; reviewer/time*; checklist confirmations*; closure rationale*; evidence/approval refs*; final residual risk/outcome*; SLA result*; limitations* if exceptional; exceptional flag; record version*.	Resolved only; authorised reviewer; monitoring complete; evidence resolvable; SoD; manager+governance for exceptional; no overdue mandatory action.	CLOSED.

Form	Fields (* required)	Validation	Audit output
F10 Suppression request	Reason code*; narrative*; scope*; requested start/expiry*; evidence*; conditions; review date*; recurrence/reactivation criteria*; prior state/owner and SLA consumed/remaining (read-only); record version*.	Open-Monitoring only; expiry future/in term; no overlap; material requester cannot self-approve; critical concurrence rules.	APPROVAL_REQUESTED only.
F11 Approval decision	Request summary/read-only; decision*; rationale*; conditions conditional; expiry conditional; evidence reviewed*; authority/SoD result read-only; record version*.	Pending/conditional request; current identity distinct; role authorised; no backdate; rejection no effect; conditional approval needs explicit conditions.	APPROVAL_* decision.
F12 Apply approved change	Approval ref*; requested effect read-only; executor*; effective time (server); condition evidence conditional; confirmation*; record version*.	Approval Approved/conditions satisfied/unexpired; target unchanged enough to apply; executor authorised; revalidate term/SoD.	APPROVED_CHANGE_APPLIED plus target event.
F13 Reopen/withdraw	Trigger/reason*; invalidating observation*; prior resolution/suppression ref*; evidence*; target state read-only; approval conditional; record version*.	Pre-close same episode; post-close successor handled by System; no SLA reset; one event only.	RESOLUTION_WITHDRAWN/REOPENED.
F14 Pipeline correction/reprocess	Run/step/source*; defect code*; affected objects/count*; correction reason*; source authority*; before/after/checksum*; admin identity*; manager authorisation conditional; new attempt reason*.	Prior run immutable; critical defects block publication; new run/attempt ID; reconciliation required before publish.	DATA_CORRECTION_RECORDED/RUN_AUTHORISED.
F15 Audit export	Scope/filter summary*; date/event range*; format*; include evidence metadata yes/no; synthetic-data confirmation*; purpose*.	Bounded authorised scope; no raw secrets/binaries; event count disclosed; large export limit enforced.	AUDIT_EXPORT_GENERATED.

Conditional resolution fields

Resolution category	Conditional fields / authority
Delivery received	Posted receipt(s), quantity/UOM/time, PO/schedule link, residual exposure; one successful validated run after posting.
Supplier commitment accepted	Commitment date/quantity/contact/reliability/revised exposure; manager if High/Critical remains; for Critical closure normally waits for receipt or accepted residual risk.
Alternative supplier arranged	Replacement order/commitment, approved supplier, quantity/date, original exposure containment; manager approval.
Substitute product approved	Planner assessment, item, equivalence/usage approval, quantity/reservation, conversion; manager for Critical.
Inventory reallocated	Transfer/allocation/reservation, from/to site, quantity, donor demand impact; manager if donor risk/critical.

Resolution category	Conditional fields / authority
Demand date changed	Authoritative demand/production/customer change, old/new date, impact; manager and demand/planner consultation.
Demand cancelled	Authoritative cancellation, quantity/site/source status; manager for Critical/partial ambiguity.
Residual risk formally accepted	Exposure/consequence/controls/owner/review/expiry/alternatives; manager plus governance for Critical/safety/production-critical.
False positive	Rule/fact analysis, no actionable exposure, rule-review reference; manager for High/Critical; independent rerun/check.
Source-data correction	Admin correction, before/after, source authority, defect code, corrected run; analyst confirms; manager High/Critical; governance if exceptional.

Appendix D. Role-permission matrix

Capability	Analyst	Planner	Manager	Admin	Governance	Read-only
View Control Tower/Queue/Detail	Yes	Yes	Yes	Read	Read	Read
Assign Open	Yes	No	Yes	No	No	No
Reassign active	Own/authority	No	Yes	No	No	No
Begin investigation	Owner	If owner	Yes	No	No	No
Add operational note/evidence	Yes	Yes	Yes	Correction only	Annotation/evidence review	No
Agree supplier action	Yes	Contribute	Yes	No	No	No
Agree inventory action	Contribute	Yes	Yes	No	No	No
Begin/record monitoring	Own action	Own inventory action	Yes	No	No	No
Resolve normal category	Within authority	Inventory categories	Yes	No	No	No
Accept residual risk	Request	Request	Execute/request	No	Concur only	No
Close normal	No	No	Yes/designated	No	Review only	No
Close exceptional	No	No	Execute after approvals	No	Mandatory concurrence	No
Request suppression/override	Yes	Yes	Yes	No	No	No
Approve material request	No	No	Yes if distinct	No	Concur if required	No

Capability	Analyst	Planner	Manager	Admin	Governance	Read-only
Apply approved control	Within action authority	Within action authority	Yes	No	No	No
View approval history	Relevant	Relevant	Yes	Read	Yes	Read
Pipeline correction/reprocess	No	No	Authorise/accountable	Execute	Review	No
Audit search/export	Own/relevant	Relevant	Yes	Admin scope	Yes	Read bounded
Audit annotation	No	No	No	Admin correction only	Yes	No
Edit/delete history or score	Never	Never	Never	Never via app	Never	Never

Appendix E. MVP acceptance criteria

RELEASE DECISION

The first Streamlit release is acceptable only when the end-to-end operational control loop, negative transitions/permissions, approval separation, stale/failure behaviour, audit reconstruction and accessibility-critical paths pass. A page that renders is not sufficient evidence.

Gate	Minimum release evidence	Pass condition
E1 Shell and honesty	Screenshots/content test for synthetic label, role simulation limitation, run freshness and score language.	No production-authentication, probability or realised-benefit claim.
E2 Queue	100-record reconciliation; 25-record stable sort; required filters/pagination/drill context; missing/override/SLA indicators.	Exact records/order; <=3s p95 target or disclosed miss.
E3 Detail/explainability	Five scenario review; hand calculations; partial receipt/correction; missing signal; historical snapshot.	Every displayed value reconciles to immutable source/run/rule/evidence.
E4 Lifecycle	Golden episode and all allowed/prohibited transitions, including rapid resolution, withdrawal, suppression/reactivation and post-close successor.	Allowed paths pass; prohibited paths produce zero mutation.
E5 Role and authority	Positive/negative action tests for all roles; direct links; identity-level self-approval; concurrence.	Every permission matches Appendix D; simulated limitation visible.
E6 SLA	Critical/High/Medium clocks; after-hours/weekend/holiday/DST; pauses; reassignment; severity change; extensions; breaches.	Exact business minutes and no unauthorised reset/erasure.
E7 Approvals	Approved, conditional, rejected, expired, withdrawn and separate-apply fixtures for each material route.	Requester cannot self-approve; rejection has zero effect; approval history complete.
E8 Human resolution/closure	All ten categories with conditional evidence; lower score/receipt promise/possible resolution; normal and exceptional closure.	Zero automatic Resolved/Closed; closure only from Resolved.
E9 Audit	Golden event ledger/export; append-only negative tests; correction links; actor/time/before-after/reason/version completeness.	100% required fields; no normal edit/delete path.
E10 Pipeline health	Successful/failed/stale run fixtures; source freshness; quarantine; control totals; atomic publication; corrected reprocess.	Failed attempt never becomes current; totals reconcile or publication is blocked.
E11 Reliability	Duplicate submit and stale-record concurrent update fixtures.	No duplicate event; stale mutation rejected with draft recovery.
E12 Accessibility	Keyboard task path; focus/error handling; contrast/status-without-colour; 200% zoom/1024px; screen-reader spot test.	No critical blocker; documented component exceptions and workaround.

Appendix F. Features that should not be implemented in the first Streamlit release

Exclude	Reason / boundary
Notifications	Email, Teams, Slack, SMS, push, retry/bounce/preferences/escalation delivery. In-app queues/banners are sufficient for MVP proof.
AI	Summaries, recommended actions, message drafts, copilots or autonomous workflow changes. They add assurance scope before the core is proven.
Autonomous decisions	Automatic Resolved/Closed, risk acceptance, approval, suppression, reprioritisation, reassignment, order change, cancellation, expedite, allocation or supplier sanction.
Production identity/security	SSO, MFA, production authentication claims, enterprise RBAC/SoD, penetration testing, identity lifecycle and admin-resistant audit.
Live integrations	ERP/EDI/WMS/supplier portal, real-time events, supplier communications and external evidence/document stores.
Bulk operational mutation	Bulk assignment, notes, action, suppression, resolution, closure or approval. Context-specific evidence and authority make these unsafe for first release.
Custom workflow/configuration builder	User-designed states, transitions, approval chains, SLA rules, holiday calendars or risk rules. Use frozen configuration and narrow fixtures.
Stored binary evidence	Confidential attachments, scanned documents, OCR and content preview. Store synthetic metadata/reference only.
Tamper-proof/legal archive	WORM storage, legal holds, certified retention, backup/restore and cryptographic non-repudiation claims.
Advanced inventory optimisation	Probabilistic stockout, dynamic safety stock, sourcing/routing optimisation, production planning and autonomous allocations.
Enterprise analytics in app	Power BI recreation, broad trend dashboards, model measures and outcome-performance analytics. Keep Streamlit operational.
Public hosting and enterprise scale	Hosted endpoint, high-concurrency claims, 24/7 observability/on-call and disaster recovery until local acceptance/security gates pass.
Mobile write workflow	Complex operational writes on phones. Provide responsive read capability where practical, but do not claim mobile-first support.
Unbounded export/search	Full-history unfiltered exports or unrestricted audit queries that harm performance or exceed the synthetic/public-data boundary.

Traceability summary

Specification area	Primary governing references
Queue and detail	BRS FR-013 to FR-016; NFR-002/008; DQ-005/007; Risk RISK-001 to RISK-007; data candidate-risk contract.
Assignment/workflow	BRS FR-017 to FR-023; Frozen Operating Model Sections 2-5; data workflow contract.
SLA	BRS SLA-001 to SLA-009 as revised by Frozen Operating Model Section 6; Australia/Melbourne calendar.
Approvals/suppression	BRS GOV-004/005, FR-020; Frozen Operating Model Section 7 and binding negative controls.
Resolution/closure	BRS FR-021/022 as revised; Frozen Operating Model Section 8; human-control boundary.
Audit/evidence	BRS GOV-001 to GOV-010, FR-024; logical data-model audit/evidence contract.
Pipeline health	BRS FR-001 to FR-005, NFR-004, DQ-001 to DQ-011, REP-007/008; publication-pointer contract.
Accessibility/security honesty	BRS SEC-001 to SEC-006; GOV-008; Charter Streamlit and synthetic-evidence limitations.

Final design decision

Freeze the first Streamlit MVP as seven user-facing pages supported by state-aware forms and one append-only event history. The application is an operational control surface, not a scoring editor, BI replacement, notification platform, AI agent or production security system. Its strongest portfolio proof is a reviewer being able to move one synthetic PO-line episode through an allowed path, observe every blocked negative path, and reconstruct source -> score -> queue -> owner -> action -> approval -> resolution -> closure without hidden mutation.